

Computer Vision

Computer vision is an area of artificial intelligence concerned with extracting useful information from images and video.

Image classification assigns an image to one or more predefined categories.

Object detection identifies objects and their locations, commonly using bounding boxes.

Image segmentation assigns labels to pixels or regions of an image.

Convolutional layers detect local visual patterns such as edges, textures, and shapes.

Data augmentation creates modified training examples such as rotated, cropped, or flipped images.

Accuracy, precision, recall, and mean average precision are examples of metrics used to evaluate computer vision models.